

2020-2021 Fall
MAT123 Midterm
30/11/2020

1. Evaluate $\lim_{x \rightarrow 0^+} (\sqrt{x})^{\ln(x+1)}$.

2. Show that the function $f(x)$ defined by

$$f(x) = \begin{cases} x \arctan \frac{1}{x}, & \text{if } x > 0 \\ 0, & \text{if } x = 0 \\ \frac{x - \cos x}{x^2}, & \text{if } x < 0 \end{cases}$$

is not continuous at the point $x = 0$.

3. Find an equation of the line which is tangent to the curve

$$\cos y^2 + xy + 1 = 0$$

at the point $\left(\sqrt{\frac{2}{\pi}}, -\sqrt{\frac{\pi}{2}}\right)$. Note that $y = f(x)$.

4. A block of ice in the shape of a cube originally having volume of 3000 cm^3 . When it is melting, the surface area is decreasing at the rate of $36 \text{ cm}^2/\text{h}$. At what rate does the length of each of its edges decrease at the time its volume is 216 cm^3 ? Assume that during melting, the block of ice maintains its cubical shape.

5. a. Using the Intermediate Value Theorem and Rolle's theorem, show that the equation $e^x + x = 0$ has only one root (Note that if this root c_1 , then $c_1 \in (-1, 0)$).

b. Determine the interval of increase, decrease, and concavity of the function $f(x) = e^x + x$. By constructing a table, sketch the graph.

6. Determine (but do not evaluate) the integral corresponding to the area of the region bounded by the curves $y = -x^2 + 1$ and $y = |x| - 1$.

7. Evaluate $\int x \ln x \, dx$.

1. Let L be the value of the limit. Since the expression is continuous for $x > 0$, we can apply the logarithm function to each side of the equation. Then, we can swap the logarithm and the limit. Use the property of logarithms afterwards.

$$L = \lim_{x \rightarrow 0^+} (\sqrt{x})^{\ln(x+1)}$$

$$\ln(L) = \ln \left[\lim_{x \rightarrow 0^+} (\sqrt{x})^{\ln(x+1)} \right] = \lim_{x \rightarrow 0^+} \ln \left[(\sqrt{x})^{\ln(x+1)} \right] = \lim_{x \rightarrow 0^+} \ln \left[(\sqrt{x})^{\ln(x+1)} \right]$$

$$\ln(L) = \lim_{x \rightarrow 0^+} [\ln(x+1) \cdot \ln(\sqrt{x})] \quad [0 \cdot \infty]$$

Make it so the limit is in the form $\frac{0}{0}$ or $\frac{\infty}{\infty}$ in order to apply the L'Hôpital's rule.

$$\ln(L) = \lim_{x \rightarrow 0^+} \left[\frac{\ln(\sqrt{x})}{\frac{1}{\ln(x+1)}} \right] \quad \left[\frac{\infty}{\infty} \right]$$

$$\stackrel{\text{L'H.}}{=} \lim_{x \rightarrow 0^+} \left[\frac{\frac{1}{\sqrt{x}} \cdot \frac{1}{2\sqrt{x}}}{-\frac{1}{\ln^2(x+1)} \cdot \frac{1}{x+1}} \right] = \lim_{x \rightarrow 0^+} \left[-\frac{\ln^2(x+1) \cdot (x+1)}{2x} \right]$$

$$= \lim_{x \rightarrow 0^+} \left[-\frac{\ln^2(x+1)}{2x} \right] \cdot \lim_{x \rightarrow 0^+} (x+1) = \lim_{x \rightarrow 0^+} \left[-\frac{\ln^2(x+1)}{2x} \right] \quad \left[\frac{0}{0} \right]$$

$$\stackrel{\text{L'H.}}{=} \lim_{x \rightarrow 0^+} \left[-\frac{2 \ln(x+1) \cdot \frac{1}{x+1}}{2} \right] = \lim_{x \rightarrow 0^+} \left[\frac{\ln(x+1)}{x+1} \right] = \frac{\ln 1}{1} = 0$$

$$\ln(L) = 0, \text{ so } \boxed{L = 1}.$$

2. $\arctan \frac{1}{x}$ takes its values on $-\frac{\pi}{2} \leq \arctan \frac{1}{x} \leq \frac{\pi}{2}$. Multiply each side by x , then we get $-\frac{x\pi}{2} \leq x \arctan \frac{1}{x} \leq \frac{x\pi}{2}$. Take the limits of each side. By the squeeze theorem, we see that the limit of $x \arctan \frac{1}{x}$ at the point $x = 0$ is 0. This means that for $f(x)$, the limit from the right side also equals 0.

$$\lim_{x \rightarrow 0} \left(-\frac{x\pi}{2} \right) \leq \lim_{x \rightarrow 0} \left(x \arctan \frac{1}{x} \right) \leq \lim_{x \rightarrow 0} \left(\frac{x\pi}{2} \right) \implies 0 \leq \lim_{x \rightarrow 0} \left(x \arctan \frac{1}{x} \right) \leq 0$$

$$\therefore \lim_{x \rightarrow 0} \left(x \arctan \frac{1}{x} \right) = 0$$

From the left side, the limit is equal to as follows.

$$\lim_{x \rightarrow 0^-} \frac{x - \cos x}{x^2} = \lim_{x \rightarrow 0^-} (x - \cos x) \cdot \lim_{x \rightarrow 0^-} \frac{1}{x^2} = -\infty$$

Continuity requires the equality of one-sided limits and the value of the function at

that point. However, the one-sided limits are not equal; $0 \neq -\infty$. Therefore, $f(x)$ is discontinuous at $x = 0$.

3. Differentiate both sides implicitly.

$$\begin{aligned}\frac{d}{dx}(\cos y^2 + xy + 1) &= \frac{d}{dx}(0) \\ -\sin y^2 \cdot 2y \frac{dy}{dx} + y + x \frac{dy}{dx} &= 0 \\ \frac{dy}{dx}(-\sin y^2 \cdot 2y + x) &= -y \\ \frac{dy}{dx} &= \frac{y}{\sin y^2 \cdot 2y - x}\end{aligned}\tag{1}$$

Evaluate (1) at the point.

$$\left. \frac{dy}{dx} \right|_{\left(\sqrt{\frac{2}{\pi}}, -\sqrt{\frac{\pi}{2}}\right)} = \frac{y}{2y \sin y^2 - x} = \frac{-\sqrt{\frac{\pi}{2}}}{2 \sin \left(\left(-\sqrt{\frac{\pi}{2}}\right)^2\right) \left(-\sqrt{\frac{\pi}{2}}\right) - \sqrt{\frac{2}{\pi}}} = \frac{\sqrt{\frac{\pi}{2}}}{\sqrt{2\pi} + \sqrt{\frac{2}{\pi}}}\tag{2}$$

Recall: $y - y_0 = m(x - x_0)$, where m is the slope. Substitute m with (2) and find the tangent line.

$$y + \sqrt{\frac{\pi}{2}} = \frac{\sqrt{\frac{\pi}{2}}}{\sqrt{2\pi} + \sqrt{\frac{2}{\pi}}} \left(x - \sqrt{\frac{2}{\pi}} \right)$$

4. Let $S(t)$, $V(t)$, $a(t)$ represent the surface area, the volume, and the length of one side of the object, respectively, as a function of time. We may write the following.

$$S(t) = 6a^2(t), \quad V(t) = a^3(t)$$

Given that at $t = t_0$, $V(t_0) = 216$, $S'(t_0) = -36$. Using the relationship with the sides,

$$\begin{aligned}V(t_0) &= a^3(t_0) = 216 \rightarrow a(t_0) = 6 \\ S'(t_0) &= 12a(t_0)a'(t_0) = -36 \\ \therefore 12 \cdot 6 \cdot a'(t_0) &= -36 \rightarrow a'(t_0) = -\frac{1}{2}\end{aligned}$$

$$a'(t_0) = -\frac{1}{2} \text{ cm/h}$$

5. a. Let $f(x) = e^x + x$. f is continuous and differentiable for all $x \in \mathbb{R}$.

$$f(-1) = e^{-1} - 1 = \frac{1}{e} - 1, \quad f(0) = e - 0 = e$$

Since $f(-1) < 0$ and $f(0) > 0$ and f is continuous on the interval $[-1, 0]$, by IVT, there is at least one point x_1 that satisfies $f(x_1) = 0$. Assume that there is another distinct root x_2 . Rolle's theorem states that if f is continuous on a particular interval with endpoints having the same function value, there exists a point c on that interval such that $f'(c) = 0$ there.

$$f'(c) = e^c + 1 \geq 1 \quad [e^c > 0]$$

This yields a contradiction. Therefore, there is *only* one root.

- b. The expression is defined $\forall x \in \mathbb{R}$. Let us find the limit at infinity and the limit at negative infinity.

$$\lim_{x \rightarrow \infty} (e^x + x) = \infty \quad \lim_{x \rightarrow -\infty} (e^x + x) = -\infty$$

There are no vertical or horizontal asymptotes. However, there is a slant asymptote. Attempt a long polynomial division and we will find that the slant asymptote is $y = x$. Verify with the following limit:

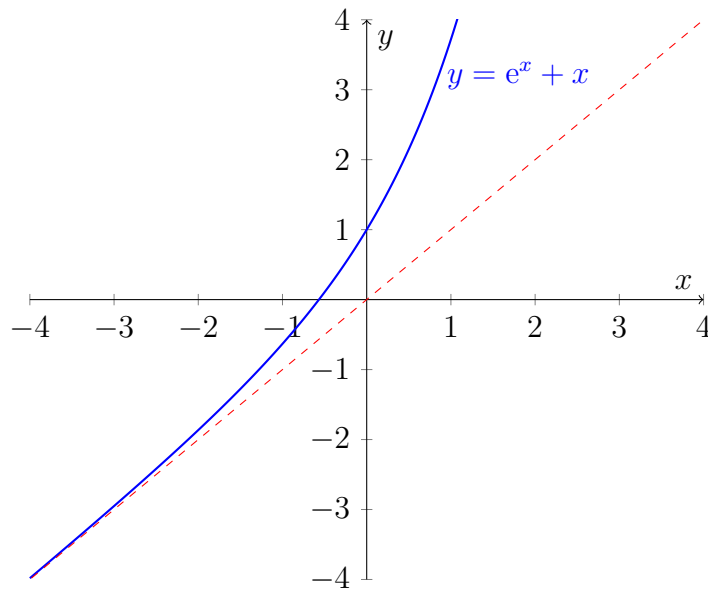
$$\lim_{x \rightarrow -\infty} [(e^x + x) - x] = \lim_{x \rightarrow -\infty} e^x = 0$$

Take the first and second derivatives.

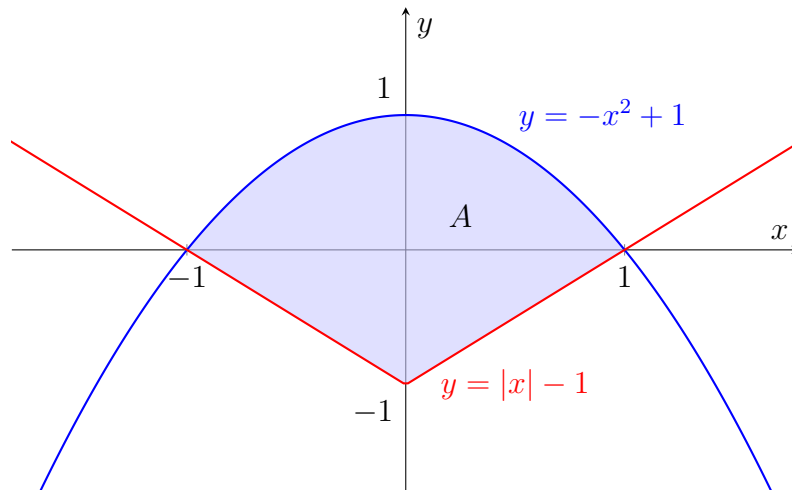
$$y' = e^x + 1, \quad y'' = e^x$$

We see that there are no critical or inflection points either. Now, set up a table and see what the graph looks like.

x	$(-\infty, \infty)$
y	$(-\infty, \infty)$
y' sign	+
y'' sign	+



6.



The area of the region is as follows.

$$A = \int_{-1}^1 [(-x^2 + 1) - (|x| - 1)] dx = \int_{-1}^0 (-x^2 + x + 2) dx + \int_0^1 (-x^2 - x + 2) dx$$

7. We'll use integration by parts.

$$\left. \begin{array}{l} \ln x = u \rightarrow \frac{1}{x} dx = du \\ x dx = dv \rightarrow \frac{x^2}{2} = v \end{array} \right\} \quad I = \int x \ln x dx = \frac{x^2}{2} \cdot \ln x - \int \frac{x^2}{2} \cdot \frac{1}{x} dx$$

$$= \frac{x^2}{2} \cdot \ln x - \int \frac{x}{2} dx = \boxed{\frac{x^2}{2} \cdot \ln x - \frac{x^2}{4} + C, C \in \mathbb{R}}$$